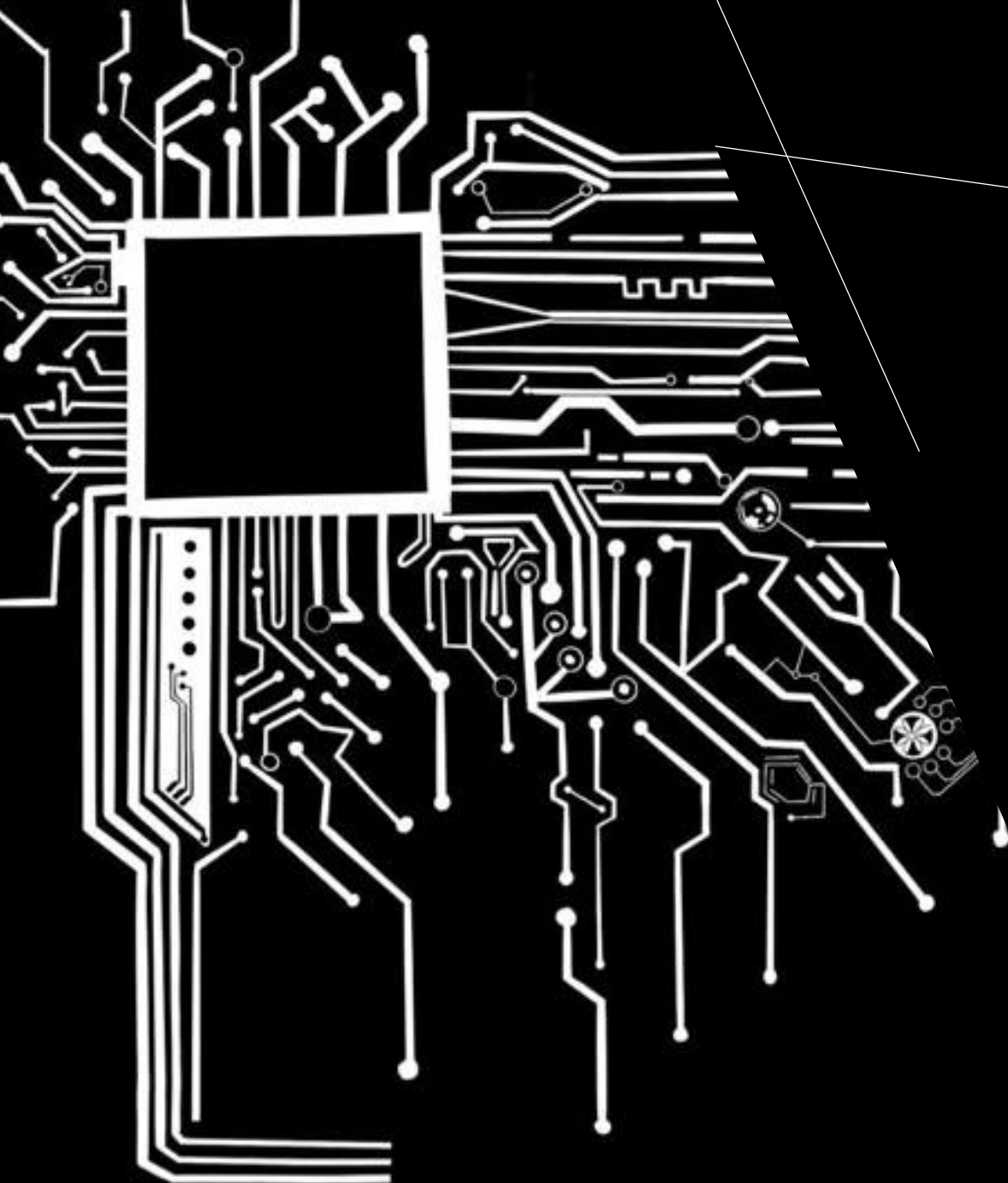




ICE 2343

DIGITAL LOGIC DESIGN

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Flip-Flop Conversion Rules

Steps of flip-flop conversion:

- i. Identify given flip-flop and the flip-flop to be designed.
- ii. Make excitation table of given or available flip-flop.
- iii. Make characteristic table of the flip-flop to be designed.
- iv. Make K-map for given flip-flop's inputs.
- v. Derive the Boolean expression for available flip-flop.
- vi. Draw the circuit for required flip-flop.

Excitation Table

Q	Q+	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

Q	Q+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Q	Q+	T
0	0	0
0	1	1
1	0	1
1	1	0

Q	Q+	D
0	0	0
0	1	1
1	0	0
1	1	1

D to T Conversion

SR excitation table

Q_n	D	Q_{n+1}
0	0	0
0	1	1
1	0	0
1	1	1

JK characteristics table

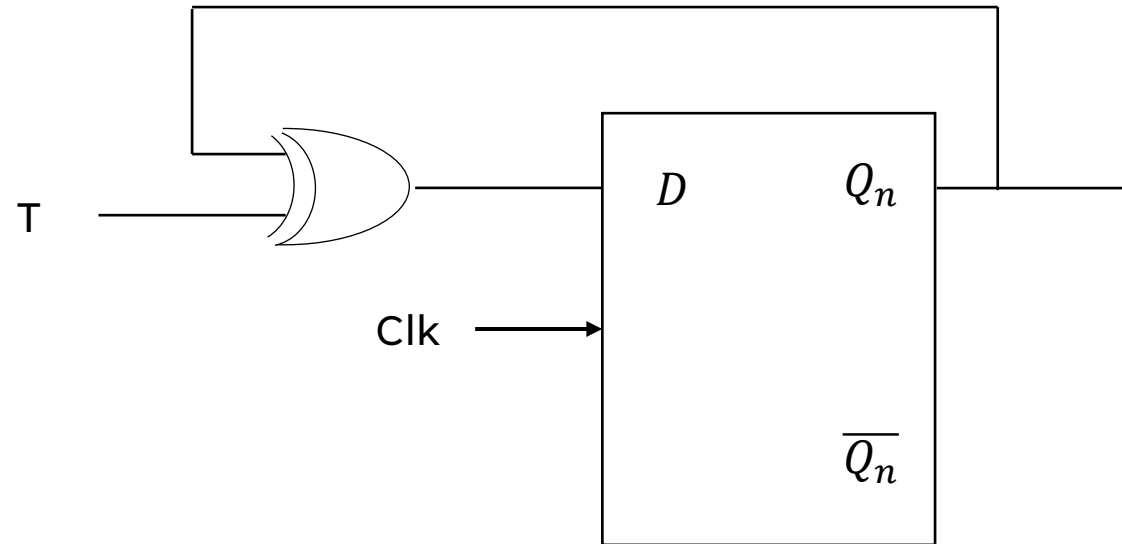
Q_n	D	Q_{n+1}	T
0	0	0	0
0	1	1	1
1	0	0	1
1	1	1	0

D to T Conversion

For D:-

T - Q_n	0	1
0		1
1	1	

$$\begin{aligned}\therefore D &= \overline{Q_n}T + \bar{T}Q_n \\ &= T \oplus Q_n\end{aligned}$$



SR to JK Conversion

SR excitation table

Q_n	Q_{n+1}	S	R
0	0	0	x
0	1	1	0
1	0	0	1
1	1	x	0

JK characteristics table

Q_n	J	K	Q_{n+1}	S	R
0	0	0	0	0	x
0	0	1	0	0	x
0	1	0	1	1	0
0	1	1	1	1	0
1	0	0	1	x	0
1	0	1	0	0	1
1	1	0	1	x	0
1	1	1	0	0	1

SR to JK Conversion

For S:-

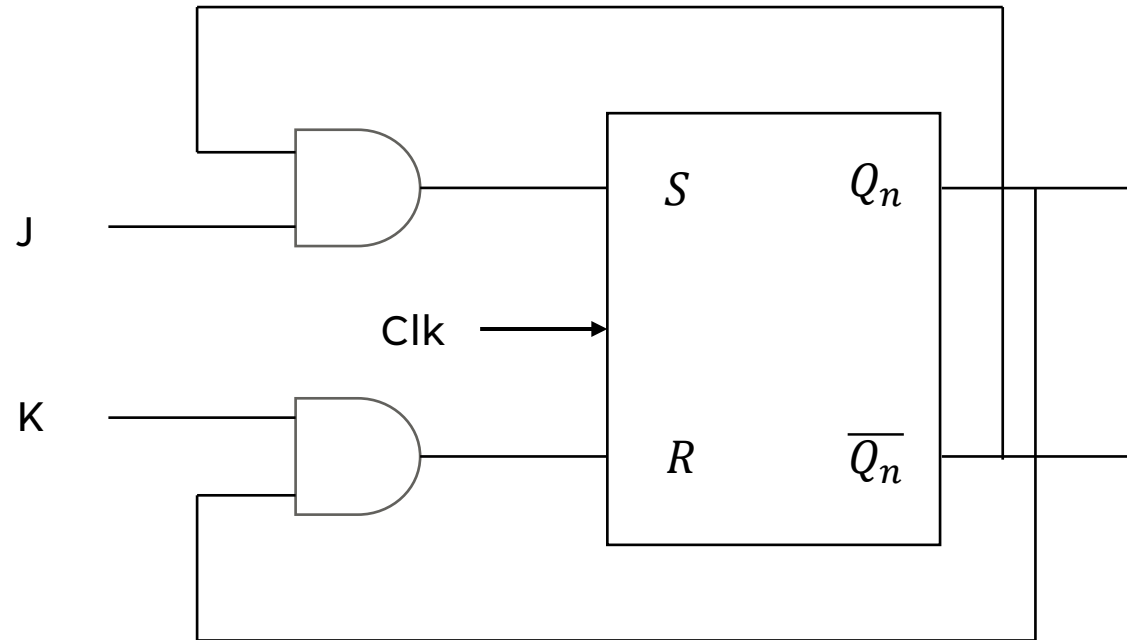
JK - Q_n	00	01	11	10
0			1	1
1	x			x

$$\therefore S = \overline{Q_n} J$$

For R:-

JK - Q_n	00	01	11	10
0	x	x		
1		1	1	

$$\therefore R = Q_n K$$



D to SR Flip-Flop

D flip-flop excitation table

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

SR characteristics table

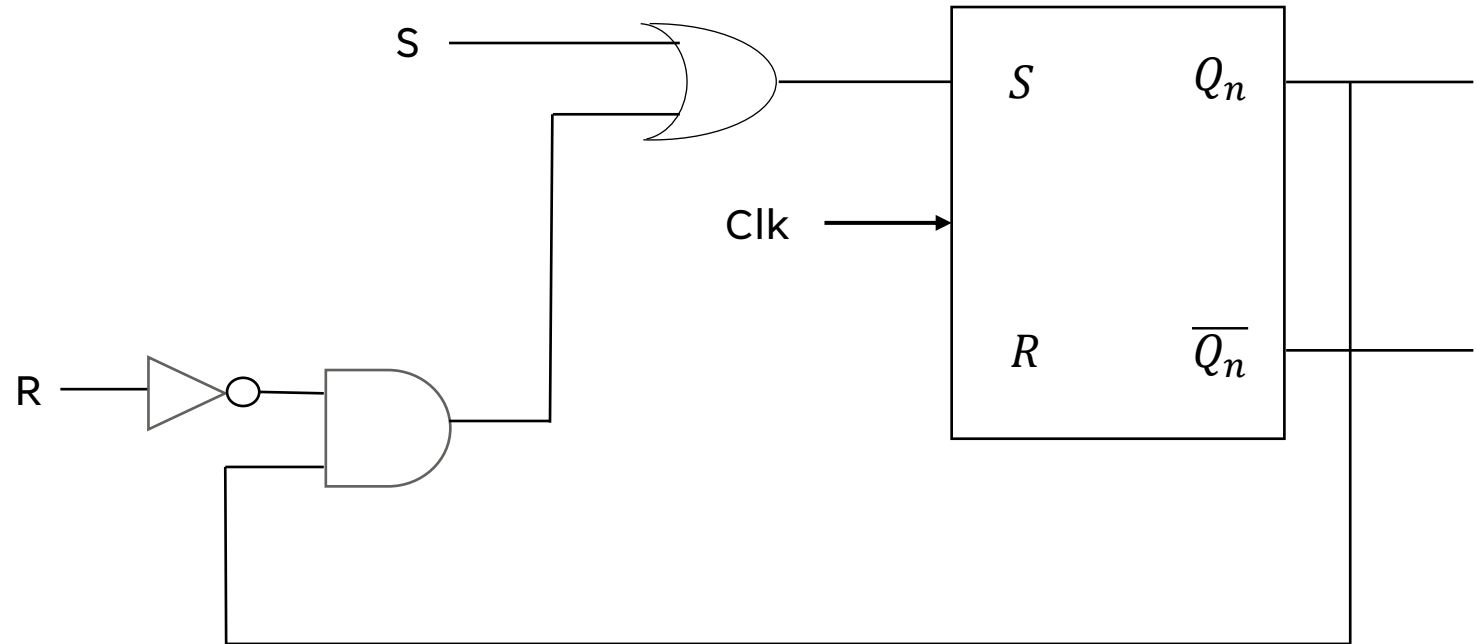
Q_n	S	R	Q_{n+1}	D
0	0	0	0	0
0	0	1	0	0
0	1	0	1	1
0	1	1	x	x
1	0	0	1	1
1	0	1	0	0
1	1	0	1	1
1	1	1	x	x

D to SR Flip-Flop

For D:-

SR - Q_n	00	01	11	10
0			x	1
1	1		x	1

$$\therefore D = S + Q_n \bar{R}$$



Exercise

- JK to SR flip-flop
- JK to D flip-flop

A series of white, thin, overlapping geometric lines on a black background, forming a complex, abstract shape on the left side of the slide.

THANK YOU